

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

EX-5004-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Power Electronic Devices and Circuits

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) What are the various methods of turn on of SCRs? Explain.
b) Give the design of snubber circuit and obtain the values of various parameters.
2. a) What is Commutation? Explain class D commutation.
b) Give the merits and demerits of GTO, as compared to conventional thyristors.
3. a) Explain the operation principle of 3 ϕ full converters with associated waveforms.
b) Explain the effect of source impedance on the performance of 3 ϕ converters.
4. a) With a neat diagram explain the working 1 ϕ full wave fully controlled bridge converter and also draw the waveform of output voltage and output current.
b) Explain the use of flywheel diode in converters with necessary output voltage waveform.

EX-5004-CBGS

PTO

[2]

5.
 - a) Explain the principle of operation of McMurray bed ford inverters.
 - b) Explain in brief the various techniques to reduction of harmonics in the inverter output voltage.
6.
 - a) Describe the working principle of step up chopper with circuit diagram and draw its output waveform.
 - b) Explain Tone's chopper with circuit diagram and give its field of applications.
7.
 - a) Explain cycloconverter operation with neat circuit diagram and waveform.
 - b) Explain the working of Buck-Boost regulators and its application.
8. Explain the following. (any two)
 - a) Series and parallel operation of SCR
 - b) Triac
 - c) Comparison of mid point and bridge rectifier circuits
 - d) Voltage source inverter

EX-5004-CBGS